

Hardware Business Management System

Purpose of This Document

Defines entities, relationships, and database design direction for MySQL and SQLite.

Known Project Scope

- Mobile application and web administration system.
- Super Admin and Shop Admin roles.
- Products, purchases, sales, inventory and stock movement.
- Expenses, capital, debts, profit reporting, and 60/40 allocation support.
- Daily opening and daily closing workflows.
- Offline-first mobile storage using SQLite and central MySQL synchronization.
- Notifications, reports, audit trail, validation, error handling, security, and acceptance criteria.

Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

Status	Meaning	How to Treat It
Confirmed	Agreed project requirement or design direction.	Use as baseline for planning and implementation.
Proposed	Recommended approach that may still change.	Validate during design and owner review.
Pending Decision	Business, accounting, or technical rule not fully approved.	Do not hard code. Ask the owner or make configurable.

Core Entity Design

Entity	Purpose	Key Fields	Status
Business	Owner business profile.	id, name, contact, settings	Confirmed

Entity	Purpose	Key Fields	Status
Shop	Branch or shop location.	id, business_id, name, location, status	Confirmed
User	System account.	id, role_id, shop_id, name, login, password_hash, status	Confirmed
Product	Hardware item.	id, category_id, code, name, unit, cost, selling_price, reorder_level	Confirmed
StockBalance	Current quantity by product and shop.	shop_id, product_id, quantity	Confirmed
StockMovement	Every stock change.	product_id, shop_id, type, qty, source, before_qty, after_qty	Confirmed
Sale and SaleItem	Sales header and sold products.	shop_id, user_id, total, payment_status, items	Confirmed
Purchase and PurchaseItem	Purchase header and purchased products.	shop_id, supplier_id, total, payment_status, items	Confirmed
Expense	Business expense.	shop_id, category_id, amount, reason, date	Confirmed
CapitalEntry	Capital movement.	shop_id, type, amount, reason	Confirmed
Debt and DebtPayment	Debt ledger and repayment history.	party, type, original_amount, balance, payments	Confirmed
DailySession	Daily opening and closing.	shop_id, business_date, opening, closing, status	Confirmed
AuditLog	Trace critical actions.	actor, action, entity, before_value, after_value	Confirmed
SyncQueue	Offline sync tracking.	local_uuid, entity, payload, status, retries, error	Confirmed

ERD Relationship Description

- Business has many Shops and Users.
- Shop has many Sales, Purchases, Expenses, Debts, Capital Entries, Daily Sessions, Stock Balances, and Stock Movements.
- Product belongs to Category and appears in Sale Items, Purchase Items, Stock Balances, and Stock Movements.
- Sale has many Sale Items; Purchase has many Purchase Items.
- Debt has many Debt Payments.
- Audit Log can reference any critical entity by entity type and record id.
- Sync Queue links local mobile operations to server records after synchronization.

Open Decisions That Must Not Be Invented

Decision Area	What Must Be Confirmed	Status
Profit formula	Exact formula for gross profit, net profit, cash profit, and expense treatment.	Pending Decision
Stock costing method	Latest cost, weighted average, FIFO, or another method for cost of goods sold.	Pending Decision
60/40 destination labels	Meaning of 60 percent and 40 percent amounts in business reports.	Pending Decision
Allocation timing	Daily, weekly, monthly, per closing, or owner-approved timing.	Pending Decision
Tax or VAT handling	Whether tax calculation is part of first release and how it affects reports.	Pending Decision
Returns refunds discounts	How returns, damaged goods, refunds, and discounts affect stock and profit.	Pending Decision

Acceptance of This Document

- The document is acceptable when the owner can identify which items are confirmed, proposed, and pending decision.
- A developer or Cloud AI can use the confirmed parts without needing to guess the project direction.
- Pending accounting rules remain visible and are not presented as final business rules.